Multiplicative functions

Student handout

A problem set on prime powers and divisor sums

8 problems | 100 points + 10 optional challenge points | Suggested time: 2–3 hours

Preparation. Unique factorization, gcds, finite sums, and inclusion–exclusion.

Name: _____ **Due:** _____

Instructions. Write on separate paper. Show factorizations and justify your answers. When asked for a proof, do not merely cite the statement. You may use results from earlier problems even if you have not proved them. Computation may check examples but does not replace a proof.

Conventions. All integers and divisors in this assignment are positive unless stated otherwise. An arithmetic function is a function on the positive integers with real values. It is *multiplicative* if $f(1) = 1$ and $f(mn) = f(m)f(n)$ whenever $\gcd(m, n) = 1$; it is *completely multiplicative* if $f(1) = 1$ and the product rule holds for all m, n. Thus $f(\prod_i p_i^{a_i}) = \prod_i f(p_i^{a_i})$ for multiplicative f and distinct primes p_i. Empty products have value 1.

I. What prime powers tell us

1. Read the definition carefully [10 points]

Suppose f is multiplicative and

$$f(2) = 2, \qquad f(4) = 5, \qquad f(3) = -1, \qquad f(9) = 4, \qquad f(5) = 0.$$

(a) Find $f(1), f(12), f(18), f(36)$, and $f(45)$. (5)

(b) Do these data determine $f(8)$? Explain by describing two multiplicative functions satisfying all the given data but having different values at 8. (3)

(c) Could f be completely multiplicative? Justify your answer. (2)

2. Which operations preserve multiplicativity? [10 points]

Let f and g be multiplicative functions.

(a) Prove that the pointwise product $h(n) = f(n)g(n)$ is multiplicative. Prove the same for $q(n) = f(n)/g(n)$ if $g(n) \neq 0$ for every n. (4)

(b) Explain why the pointwise sum $f + g$ is never multiplicative under our convention. (2)

(c) Normalizing the sum fixes its value at 1. Does it follow that $A(n) = (f(n) + g(n))/2$ is always multiplicative? Prove this or give an explicit counterexample with coprime inputs. (4)

3. Counting and adding divisors [14 points]

Define $\tau(n) = \sum_{d|n} 1$ and $\sigma(n) = \sum_{d|n} d$. Write $n = \prod_{i=1}^{r} p_i^{a_i}$, with distinct primes p_i.

(a) Describe the divisors of n using exponents. Deduce product formulas for $\tau(n)$ and $\sigma(n)$, and explain why both functions are multiplicative. (6)

(b) Compute $\tau(540)$ and $\sigma(540)$. (4)

(c) Prove that $\tau(n)$ is odd if and only if n is a perfect square. Include $n = 1$. (4)

II. Totients and divisor sums

4. The totient from excluded multiples [14 points]

Let $\varphi(n)$ count integers $1 \le k \le n$ with $\gcd(k, n) = 1$; in particular, $\varphi(1) = 1$.

(a) For a prime p and $a \ge 1$, prove $\varphi(p^a) = p^a - p^{a-1}$. (3)

(b) Apply inclusion–exclusion to the multiples of the distinct primes dividing n to prove

$$\varphi(n) = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right).$$

Use this formula to prove that φ is multiplicative. (6)

(c) Compute $\varphi(540)$. (2)

(d) Prove that $\varphi(n)$ is even for every $n > 2$. *Hint:* pair k with $n - k$. (3)

5. Multiplication by a divisor sum [14 points]

The *Dirichlet convolution* of arithmetic functions is

$$(f * g)(n) = \sum_{d \mid n} f(d) g(n/d).$$

Let $\mathbf{1}(n) = 1$, $\mathrm{id}(n) = n$, and let $\varepsilon(1) = 1$, $\varepsilon(n) = 0$ for $n > 1$.

(a) Suppose $\gcd(m, n) = 1$. Prove that every divisor of mn has a unique expression ab with $a \mid m$ and $b \mid n$. (4)

(b) Use (a) to prove that $f * g$ is multiplicative when f and g are multiplicative. Check the value at 1 and explain each use of coprimality. (6)

(c) Identify $\mathbf{1} * \mathbf{1}$ and $\mathbf{1} * \mathrm{id}$ in terms of the functions already studied. For an arbitrary arithmetic function f, prove $f * \varepsilon = f$. (4)

6. Möbius cancellation [14 points]

An integer is *squarefree* if no prime square divides it. Define

$$\mu(n) = \begin{cases} 1, & n = 1, \\ (-1)^r, & n \text{ is a product of } r \text{ distinct primes}, \\ 0, & n \text{ is not squarefree}. \end{cases}$$

(a) Prove that μ is multiplicative. Show that it is not completely multiplicative. (3)

(b) Find $\sum_{d \mid p^a} \mu(d)$ for a prime p and $a \ge 1$. Use Problem 5 to prove $\mu * \mathbf{1} = \varepsilon$, including its value at 1. (5)

(c) You may use associativity and commutativity of convolution. Prove the following identities for every $n \ge 1$. (6)

$$\sum_{d \mid n} \mu(d) \sigma(n/d) = n, \qquad \sum_{d \mid n} \mu(d) \tau(n/d) = 1.$$

III. Two applications

7. An alternating divisor sum detects squares [12 points]

For $n = \prod_{i=1}^{r} p_i^{a_i}$ define $\Omega(n) = a_1 + \cdots + a_r$, with $\Omega(1) = 0$, and set $\lambda(n) = (-1)^{\Omega(n)}$. Thus Ω counts prime factors *with multiplicity*.

(a) Prove that λ is completely multiplicative. (4)

(b) Evaluate $\sum_{j=0}^{a} \lambda(p^j)$ and deduce, for every n,

$$\sum_{d \mid n} \lambda(d) = \begin{cases} 1, & n \text{ is a perfect square,} \\ 0, & n \text{ is not a perfect square.} \end{cases}$$

Justify the passage from prime powers to general n, and include $n = 1$. (6)

(c) Evaluate $\sum_{d \mid 360} \lambda(d)$ and $\sum_{d \mid 900} \lambda(d)$ without listing all the divisors. (2)

8. A sum of gcds [12 points]

Define $G(n) = \sum_{k=1}^{n} \gcd(k, n)$.

(a) Group the integers k according to $d = \gcd(k, n)$ to prove

$$G(n) = \sum_{d \mid n} d \, \varphi(n/d).$$

Explain why the group with gcd equal to d has $\varphi(n/d)$ members. (5)

(b) Use the earlier problems to deduce that G is multiplicative. (2)

(c) For a prime p and $a \geq 1$, derive

$$G(p^a) = (a+1)p^a - a p^{a-1}.$$

Then compute $G(60)$ without summing 60 gcds individually. (5)

Optional challenge: Even perfect numbers [10 extra points]

An integer N is *perfect* if its positive divisors other than N sum to N, equivalently $\sigma(N) = 2N$.

(a) Let $r \geq 2$ and suppose $2^r - 1$ is prime. Prove that $N = 2^{r-1}(2^r - 1)$ is perfect. Give the examples from $r = 2, 3, 5$. (4)

(b) Prove that every even perfect number has the form in (a). *Hint:* write $N = 2^a m$ with $a \geq 1$ and m odd, and put $q = 2^{a+1} - 1$. First show $q \mid m$. Write $m = qt$ and show $\sigma(m) = m + t$. If $t > 1$, compare this with the contribution of the three distinct divisors $1, t, m$. (6)